

Week 11: Censored Models and Sample Selection

Samuel Rowe adapted from Wooldridge (2011), Mitchell (2020), and UCLA OARC (2025)

2025-11-11

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bookdown::render_book()

Overview

This week we will be covering:

1. Censored Regression Models
2. Truncated Regression Models
3. Sample Selection
4. Subgroup Functions Part II

Chapter 1

Censored Models

1.1 Censored Regression Model - Right-Censoring

Lesson: We can use a Tobit estimator for right-censored model

We need to summarize the dependent variable to see the right-censored value. We know that weekly earnings in the Current Population Survey are top-coded or right-censored at \$2884.61. This may bias our estimate, so we'll compare a OLS model and a Tobit model for right-censored data. Remember a Tobit estimator for a censored model is different from a corner solution.

```
use "/Users/Sam/Desktop/Econ 645/Data/CPS/jan2024.dta", clear
sum earnings if prrelg==1, detail
histogram earnings if prrelg==1, normal title(Histogram of Weekly Earnings) caption("Source: Current Population Survey")
graph export "/Users/Sam/Desktop/Econ 645/R Markdown/week9_histogram_earnings.png", replace
```

Weekly Earnings: pternwa					

	Percentiles	Smallest			
1%	70	0			
5%	225	0			
10%	360	0	Obs		10,666
25%	656	0	Sum of Wgt.		10,666
50%	1000		Mean		1230.474
		Largest	Std. Dev.		788.1457
75%	1680	2884.61			
90%	2692.3	2884.61	Variance		621173.7

```

95%      2884.61      2884.61      Skewness      .7779644
99%      2884.61      2884.61      Kurtosis      2.648218

```

```
(bin=40, start=0, width=72.11525)
```

```
(file /Users/Sam/Desktop/Econ 645/R Markdown/week9_histogram_earnings.png written in PL
```

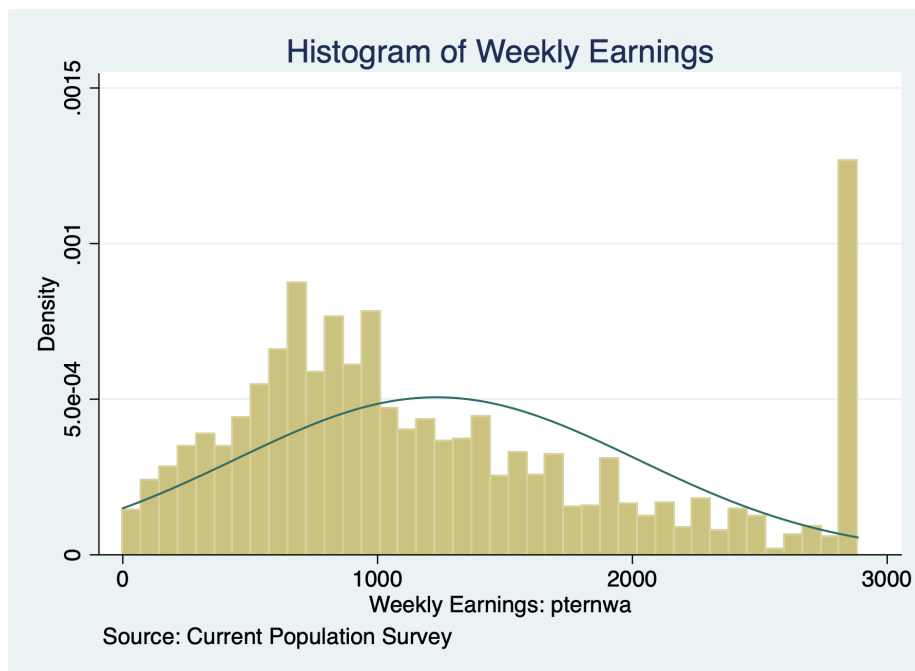


Figure 1.1: Histogram of Earnings

Let us take a look at the natural log of earnings

```

use "/Users/Sam/Desktop/Econ 645/Data/CPS/jan2024.dta", clear
sum llearnings, detail
histogram llearnings if prrelg==1, normal title(Histogram of LN Weekly Earnings) capt
graph export "/Users/Sam/Desktop/Econ 645/R Markdown/week9_histogram_llearnings.png", r

```

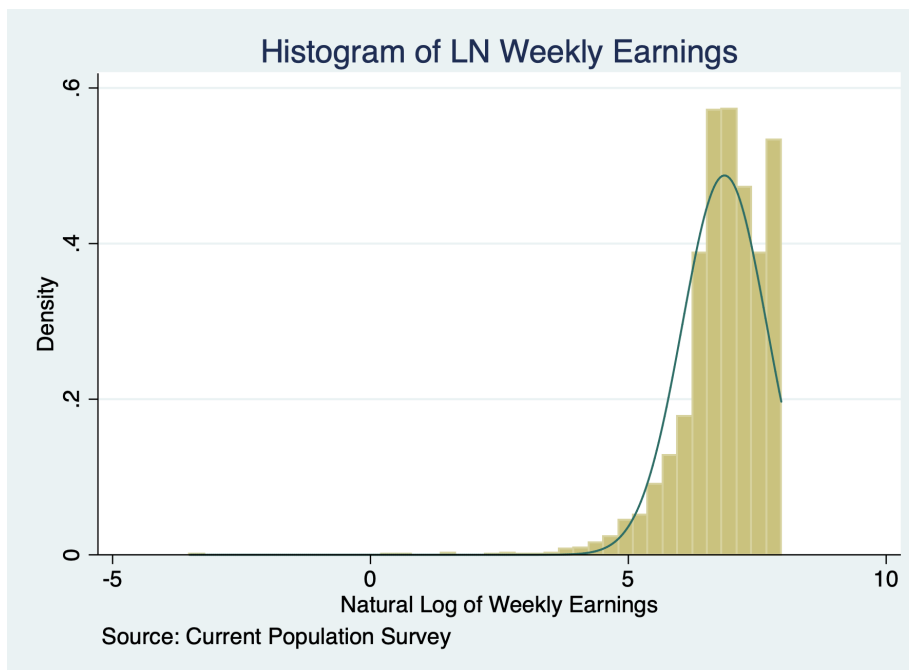
Natural Log of Weekly Earnings

	Percentiles	Smallest		
1%	4.356709	-3.506558		
5%	5.420535	-3.506558		
10%	5.886104	-3.506558	Obs	10,652
25%	6.49224	.48858	Sum of Wgt.	10,652

50%	6.907755		Mean	6.86502
		Largest	Std. Dev.	.8181897
75%	7.426549	7.967145		
90%	7.898151	7.967145	Variance	.6694343
95%	7.967145	7.967145	Skewness	-1.791008
99%	7.967145	7.967145	Kurtosis	13.92768

(bin=40, start=-3.5065579, width=.28684257)

(file /Users/Sam/Desktop/Econ 645/R Markdown/week9_histogram_lnearnings.png written in PNG format)



We'll estimate the following Mincer Equation.

$$\ln(wages_i) = \beta_0 + \beta_1 edu_i + \beta_2 exp + \beta_3 exp^2 + \beta_4 marital_i + \beta_5 veteran_i + \beta_6 union_i + \beta_7 female_i + \beta_8 race_i + u_i$$

We'll need to use the option, *ul(right-censored-value)* with our Tobit estimator.

```
est clear
eststo OLS: quietly reg lnearnings i.edu exp expsq i.marital i.veteran i.union i.female i.race
eststo Tobit: quietly tobit lnearnings i.edu exp expsq i.marital i.veteran i.union i.female i.rac
esttab OLS Tobit, drop(0.* 1.race* 1.mar* 1.edu) mtitle("OLS" "Tobit")
```

Natural Log of Weekly Earnings

	Percentiles	Smallest		
1%	4.356709	-3.506558		
5%	5.420535	-3.506558		
10%	5.886104	-3.506558	Obs	10,652
25%	6.49224	.48858	Sum of Wgt.	10,652
50%	6.907755		Mean	6.86502
		Largest	Std. Dev.	.8181897
75%	7.426549	7.967145		
90%	7.898151	7.967145	Variance	.6694343
95%	7.967145	7.967145	Skewness	-1.791008
99%	7.967145	7.967145	Kurtosis	13.92768

scalars:

```

      r(N) = 10652
    r(sum_w) = 10652
    r(mean) = 6.865020483352663
    r(Var) = .6694343494891623
    r(sd) = .8181896781854207
  r(skewness) = -1.791007993978133
  r(kurtosis) = 13.92768307319675
    r(sum) = 73126.19818867257
    r(min) = -3.506557897319982
    r(max) = 7.967144987828557
    r(p1) = 4.356708826689592
    r(p5) = 5.420534999272286
    r(p10) = 5.886104031450156
    r(p25) = 6.492239835020471
    r(p50) = 6.907755278982137
    r(p75) = 7.426549072397305
    r(p90) = 7.898151125863075
    r(p95) = 7.967144987828557
    r(p99) = 7.967144987828557

```

(1)	(2)
OLS	Tobit

main

2.edu	0.330*** (11.77)	0.330*** (11.04)
3.edu	0.442*** (13.48)	0.445*** (12.67)
4.edu	0.788*** (26.50)	0.832*** (26.11)
5.edu	0.951*** (30.01)	1.046*** (30.64)
exp	0.0558*** (28.73)	0.0575*** (27.60)
expsq	-0.000887*** (-28.27)	-0.000911*** (-27.03)
2.marital	-0.0389 (-1.92)	-0.0462* (-2.12)
3.marital	-0.118*** (-6.66)	-0.130*** (-6.84)
1.veteran	0.0412 (1.34)	0.0428 (1.28)
1.union	0.0670** (3.05)	0.0448 (1.90)
1.female	-0.304*** (-22.81)	-0.335*** (-23.25)
2.race_eth~y	0.0325 (0.41)	0.0551 (0.65)
3.race_eth~y	-0.0466 (-0.60)	-0.0576 (-0.69)
4.race_eth~y	0.135 (1.05)	0.131 (0.96)
5.race_eth~y	0.0232 (0.30)	0.0167 (0.20)
6.race_eth~y	0.0730	0.0903

	(0.82)	(0.95)
7.race_eth~y	0.0549 (0.73)	0.0553 (0.69)
_cons	5.813*** (69.47)	5.817*** (64.82)

sigma		
_cons		0.714*** (137.00)

N	10568	10568

t statistics in parentheses		
* p<0.05, ** p<0.01, *** p<0.001		

Interpretation A very similar interpretation to OLS, but we need normality and homoskedasticity for unbiased estimator. We can use a log-linear interpretation without a scaling factor, so the returns to education would only require $(e^\beta - 1) * 100$ interpretation.

1.2 Censored Regression Model - Duration analysis

Lesson: We can look at a censored data for duration analysis similar to Wooldridge's example.

This differs from a top-coded data, which we can use a Tobit analysis that we just saw.

We can look at the duration of time in months between an arrests for inmates in a North Carolina prison after being released from prison. We want to evaluate a work program to see if it is effective in increasing duration before recidivism occurs.

Note: 893 inmates have not been arrested during the period they were followed. These observations are censored. The censoring times differed among inmates ranging from 70 to 81 months.

Our dependent variable duration (time in months) is transformed by natural logarithm. We have a bunch of observations that recidivate between 70 and 81 months.

```
use "/Users/Sam/Desktop/Econ 645/Data/Wooldridge/recid.dta", clear
tab durat
```

We have a bunch of observations that are censored after 69 months (but not all)

```
use "/Users/Sam/Desktop/Econ 645/Data/Wooldridge/recid.dta", clear
tab durat cens
```

	durat	cens		Total
		0	1	
1		8	0	8
2		15	0	15
3		14	0	14
4		13	0	13
5		16	0	16
6		18	0	18
7		18	0	18
8		16	0	16
9		18	0	18
10		22	0	22
11		11	0	11
12		14	0	14
13		15	0	15
14		16	0	16
15		23	0	23
16		11	0	11
17		9	0	9
18		16	0	16
19		9	0	9
20		8	0	8
21		13	0	13
22		7	0	7
23		16	0	16
24		12	0	12
25		13	0	13
26		8	0	8
27		11	0	11
28		9	0	9
29		8	0	8
30		7	0	7
31		6	0	6
32		6	0	6
33		6	0	6
34		4	0	4

35	5	0	5
36	6	0	6
37	6	0	6
38	4	0	4
39	4	0	4
40	2	0	2
41	7	0	7
42	5	0	5
43	5	0	5
44	4	0	4
45	4	0	4
46	7	0	7
47	4	0	4
48	1	0	1
49	4	0	4
50	5	0	5
51	2	0	2
52	2	0	2
53	8	0	8
54	3	0	3
55	5	0	5
56	2	0	2
57	4	0	4
58	1	0	1
59	5	0	5
60	3	0	3
62	3	0	3
63	2	0	2
64	1	0	1
65	2	0	2
66	3	0	3
67	3	0	3
68	4	0	4
69	2	0	2
70	2	103	105
71	2	88	90
72	1	84	85
73	1	107	108
74	1	71	72
75	0	44	44
76	0	105	105
77	1	60	61
78	0	54	54
79	0	60	60
80	0	69	69
81	0	48	48

```
-----+-----+-----
      Total |      552      893 |      1,445
```

We'll use the *stset* command and set failure at `cens==0`. We'll use the *stset* command to time set survival.

```
use "/Users/Sam/Desktop/Econ 645/Data/Wooldridge/recid.dta", clear
stset ldurat, failure(cens==0)
```

```
      failure event:  cens == 0
obs. time interval:  (0, ldurat]
      exit on or before:  failure
```

```
-----
      1445  total observations
           8  observations end on or before enter()
-----
      1437  observations remaining, representing
           544  failures in single-record/single-failure data
      5411.742  total analysis time at risk and under observation
                                     at risk from t =           0
                        earliest observed entry t =           0
                                     last observed exit t =  4.394449
```

We'll use the *streg* command and set the distribution

$$ldurat_i = \alpha + \delta workprg_i + \beta_2 tserved_i + \beta_3 felon_i + \beta_4 alcohol_i + \beta_5 drugs_i + \beta_6 educ_i + x_i' \gamma + \varepsilon_i$$

Where x' are demographics of race, marital status, and age.

```
streg workprg priors tserved felon alcohol drugs black married educ age, dist(weibull) nohr
```

```
      failure event:  cens == 0
obs. time interval:  (0, ldurat]
      exit on or before:  failure
```

```
-----
      1445  total observations
           8  observations end on or before enter()
-----
      1437  observations remaining, representing
           544  failures in single-record/single-failure data
      5411.742  total analysis time at risk and under observation
```

```

                                at risk from t =          0
    earliest observed entry t =          0
                                last observed exit t =  4.394449

```

```

    failure _d:  cens == 0
analysis time _t:  ldurat

```

Fitting constant-only model:

```

Iteration 0:  log likelihood = -1254.5107
Iteration 1:  log likelihood = -1100.1536
Iteration 2:  log likelihood = -1079.1128
Iteration 3:  log likelihood = -1078.7957
Iteration 4:  log likelihood = -1078.7957

```

Fitting full model:

```

Iteration 0:  log likelihood = -1078.7957
Iteration 1:  log likelihood = -1034.1821
Iteration 2:  log likelihood = -1001.9186
Iteration 3:  log likelihood = -1000.5996
Iteration 4:  log likelihood = -1000.5919
Iteration 5:  log likelihood = -1000.5919

```

Weibull regression -- log relative-hazard form

```

No. of subjects =          1,437          Number of obs   =          1,437
No. of failures =           544
Time at risk    =  5411.742317
Log likelihood  =  -1000.5919
LR chi2(10)     =          156.41
Prob > chi2     =          0.0000

```

_t	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
workprg	.0780722	.091463	0.85	0.393	-.1011921	.2573364
priors	.0834203	.0139155	5.99	0.000	.0561465	.1106941
tserve	.0134545	.0016851	7.98	0.000	.0101519	.0167572
felon	-.287139	.106697	-2.69	0.007	-.4962614	-.0780167
alcohol	.4399456	.10658	4.13	0.000	.2310527	.6488385
drugs	.2920932	.0983695	2.97	0.003	.0992926	.4848938
black	.4515388	.0889883	5.07	0.000	.2771249	.6259527
married	-.146192	.1098131	-1.33	0.183	-.3614216	.0690377
educ	-.023948	.019578	-1.22	0.221	-.0623202	.0144241
age	-.0036431	.0005284	-6.89	0.000	-.0046788	-.0026073
_cons	-3.639277	.3077568	-11.83	0.000	-4.24247	-3.036085

/ln_p		.9214587	.0396737	23.23	0.000	.8436997	.9992178
p		2.512953	.0996982			2.324953	2.716156
1/p		.3979381	.0157877			.3681673	.4301163

Interpretation: Given the log-linear function form, we can easily determine the estimated percent change in duration before criminal recidivism.

```
display (exp(.0780722)-1)*100
```

8.1200719

Being a part of the work program increase the duration of time before recidivism, but it is not statistically significant.

```
display (exp(-.287139)-1)*100
```

-24.959259

Being a felon reduces the duration of time before recidivism, where a felon has as 24% decrease in duration of time before recidivism.

Chapter 2

Truncated Regression Models

We will again look at the January 2024 Current Population Survey data. However, we will truncate the data at \$2884 per week by dropping all observation at or above this threshold.

2.1 Truncated Data

We will set the threshold $c_i \geq 2884$. Every observation at or above this threshold will be dropped.

```
use "/Users/Sam/Desktop/Econ 645/Data/CPS/jan2024.dta", clear
sum earnings if prerelg==1, detail
est clear
eststo OLS: quietly reg lnearnings i.edu exp exsq i.marital i.veteran i.union i.female i.race if
drop if earnings >= 2884
sum earnings if prerelg==1, detail
```

Weekly Earnings: pternwa				

	Percentiles	Smallest		
1%	70	0		
5%	225	0		
10%	360	0	Obs	10,666
25%	656	0	Sum of Wgt.	10,666
50%	1000		Mean	1230.474

		Largest	Std. Dev.	788.1457
75%	1680	2884.61		
90%	2692.3	2884.61	Variance	621173.7
95%	2884.61	2884.61	Skewness	.7779644
99%	2884.61	2884.61	Kurtosis	2.648218

(28,566 observations deleted)

Weekly Earnings: pternwa				

	Percentiles	Smallest		
1%	62.5	0		
5%	206	0		
10%	336	0	Obs	9,767
25%	620	0	Sum of Wgt.	9,767
50%	960		Mean	1078.221
		Largest	Std. Dev.	635.0599
75%	1450	2880		
90%	2000	2880	Variance	403301.1
95%	2320	2880	Skewness	.7302792
99%	2780	2880	Kurtosis	2.975473

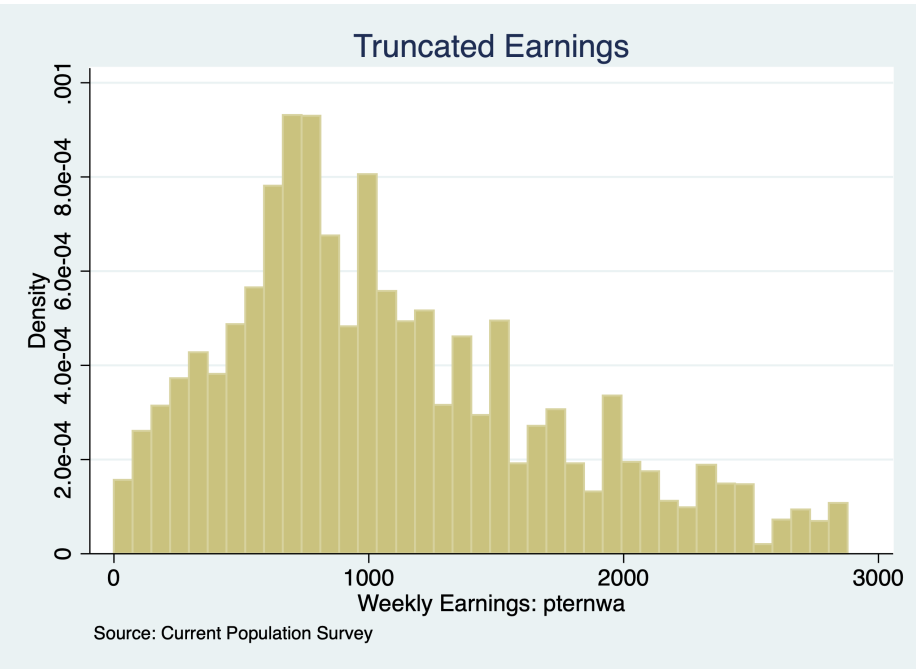
Our largest value is now \$2880 weekly earnings, which is just below the threshold.

Let's look at a histogram of the truncated weekly earnings.

```

histogram earnings if prerelg, title("Truncated Earnings") note("Source: Current Popul
graph export "/Users/Sam/Desktop/Econ 645/Data/CPS/jan2024_trunc.png", replace

```



We no longer have a spike in density at 2884 like we did with the censored data.

2.2 Truncated Regression Models

Next, we will use the command `truncreg` and set the option `ul()` at 2884.

```
eststo Truncated: truncreg llearnings i.edu exp expsq i.marital i.veteran i.union i.female i.race
```

(note: 0 obs. truncated)

Fitting full model:

```
Iteration 0: log likelihood = -9654.7635
Iteration 1: log likelihood = -9654.7551
Iteration 2: log likelihood = -9654.7551
```

Truncated regression

Limit:	lower =	-inf	Number of obs =	=	9,669
	upper =	2884	Wald chi2(17)	=	3276.34
Log likelihood =	-9654.7551		Prob > chi2	=	0.0000

	llearnings	Coef.	Std. Err.	z	P> z	[95% Conf. I
--	------------	-------	-----------	---	------	--------------

-----+-----					
	edu				
	HS/GED		.3230896	.0276751	11.67 0.000
	AA		.4320131	.0325527	13.27 0.000
	BS/BA		.6971765	.0297669	23.42 0.000
	AdDegree		.8088922	.0325213	24.87 0.000
	exp		.054815	.0019609	27.95 0.000
	expsq		-.0008866	.0000319	-27.81 0.000
	marital				
	Divorced/Separated/Widowed		-.0290168	.0206418	-1.41 0.160
	Never Married		-.1028859	.0178986	-5.75 0.000
	veteran				
	Veteran		.0280901	.0322307	0.87 0.383
	union				
	Union		.1126058	.0223161	5.05 0.000
	female				
	Female		-.2718903	.0137518	-19.77 0.000
	race_ethnicity				
	NH Asian		-.0426197	.0795044	-0.54 0.592
	NH Black		-.04673	.0774491	-0.60 0.546
	NH Native Hawaiian or Pacific Islander		.136987	.1278521	1.07 0.284
	Latino/a or Hispanic		.0151638	.076424	0.20 0.843
	NH Multiracial		.0275207	.0898433	0.31 0.759
	NH White		.035842	.0750013	0.48 0.633
	_cons		5.814378	.0837643	69.41 0.000
-----+-----					
	/sigma		.6567763	.0047229	139.06 0.000
-----+-----					

We can directly compare our OLS and Truncated Regression Model coefficients.

```
esttab OLS Truncated
```

	(1)	(2)
	OLS	TRM

main		
1.edu	0	0

	(.)	(.)
2.edu	0.330*** (11.77)	0.323*** (11.67)
3.edu	0.442*** (13.48)	0.432*** (13.27)
4.edu	0.788*** (26.50)	0.697*** (23.42)
5.edu	0.951*** (30.01)	0.809*** (24.87)
exp	0.0558*** (28.73)	0.0548*** (27.95)
expsq	-0.000887*** (-28.27)	-0.000887*** (-27.81)
1.marital	0 (.)	0 (.)
2.marital	-0.0389 (-1.92)	-0.0290 (-1.41)
3.marital	-0.118*** (-6.66)	-0.103*** (-5.75)
0.veteran	0 (.)	0 (.)
1.veteran	0.0412 (1.34)	0.0281 (0.87)
0.union	0 (.)	0 (.)
1.union	0.0670** (3.05)	0.113*** (5.05)
0.female	0 (.)	0 (.)
1.female	-0.304*** (-22.81)	-0.272*** (-19.77)

1.race_eth~y	0 (.)	0 (.)
2.race_eth~y	0.0325 (0.41)	-0.0426 (-0.54)
3.race_eth~y	-0.0466 (-0.60)	-0.0467 (-0.60)
4.race_eth~y	0.135 (1.05)	0.137 (1.07)
5.race_eth~y	0.0232 (0.30)	0.0152 (0.20)
6.race_eth~y	0.0730 (0.82)	0.0275 (0.31)
7.race_eth~y	0.0549 (0.73)	0.0358 (0.48)
_cons	5.813*** (69.47)	5.814*** (69.41)

sigma		
_cons		0.657*** (139.06)

N	10568	9669

t statistics in parentheses		
* p<0.05, ** p<0.01, *** p<0.001		

Interpretation: Our union wage premium is estimated to be $((e^{0.067}) - 1) * 100\% = 6.9\%$ with the OLS estimator, while our union wage premium is estimated to be $((e^{0.113}) - 1) * 100\% = 12.0\%$.

We can see that our OLS estimates are biased upwards for education and experience, but downward biased for union wage premium.

Source: <https://stats.oarc.ucla.edu/stata/output/truncated-regression/>

Chapter 3

Sample Selection Correction

Lesson: Similar to tobit, when we don't account for the truncation of the data, or why certain parts of the population are not sampled, we will commit a type of omitted variable bias without our $\lambda(z)$ -hat. We can use a Heckit method for sample selection correction.

We want to see if there is sample selection bias due to unobservable wage offers for non-working women.

1. We need to estimate a logit or probit to test and correct for sample selection bias due to unobserved wage offer for nonworking women. Spousal income, education, experience, age, and number of kids less than 6.

$$\ln(wages_i) = \beta_0 + \mathbf{x}' + u_i$$

Where $\mathbf{x}$ is a vector that includes education, experience, and experience squared

2. We use the Heckit command to implement a Heckman Method for sample selection correction.

$$\ln(wages_i) = \beta_0 + \mathbf{x}' + \mathbf{z}' + u_i$$

Where $\mathbf{z}$ is a vector that includes spousal income, education, experience, age, and number of kids less than 6.

3.1 OLS

```
use "/Users/Sam/Desktop/Econ 645/Data/Wooldridge/mroz.dta", clear
reg lwage educ exper expersq
```

Source	SS	df	MS	Number of obs	=	428
Model	35.0222967	3	11.6740989	F(3, 424)	=	26.29
Residual	188.305144	424	.444115906	Prob > F	=	0.0000
				R-squared	=	0.1568
				Adj R-squared	=	0.1509
Total	223.327441	427	.523015084	Root MSE	=	.66642

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
educ	.1074896	.0141465	7.60	0.000	.0796837 .1352956
exper	.0415665	.0131752	3.15	0.002	.0156697 .0674633
expersq	-.0008112	.0003932	-2.06	0.040	-.0015841 -.0000382
_cons	-.5220406	.1986321	-2.63	0.009	-.9124667 -.1316144

Notice: an assumption is used to exclude spousal income, age, kids less than 6, and kids greater than 6 from our main regression.

```
display (exp(_b[educ])-1)*100
```

```
11.347933
```

3.2 Heckman Method - Heckit

We will use a subset of all exogenous variable 1. What are the factors correlated with being in the labor force? 2. What is the impact of education and experience on wages

```
use "/Users/Sam/Desktop/Econ 645/Data/Wooldridge/mroz.dta", clear
heckman lwage educ exper expersq, select(inlf=nwifeinc educ exper expersq age kidslt6 l
```

Heckman selection model -- two-step estimates	Number of obs	=	753
(regression model with sample selection)	Censored obs	=	325
	Uncensored obs	=	428

Wald chi2(3) = 51.53
 Prob > chi2 = 0.0000

	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
lwage						
educ	.1090655	.015523	7.03	0.000	.0786411	.13949
exper	.0438873	.0162611	2.70	0.007	.0120163	.0757584
expersq	-.0008591	.0004389	-1.96	0.050	-.0017194	1.15e-06
_cons	-.5781032	.3050062	-1.90	0.058	-1.175904	.019698
inlf						
nwifeinc	-.0120237	.0048398	-2.48	0.013	-.0215096	-.0025378
educ	.1309047	.0252542	5.18	0.000	.0814074	.180402
exper	.1233476	.0187164	6.59	0.000	.0866641	.1600311
expersq	-.0018871	.0006	-3.15	0.002	-.003063	-.0007111
age	-.0528527	.0084772	-6.23	0.000	-.0694678	-.0362376
kidslt6	-.8683285	.1185223	-7.33	0.000	-1.100628	-.636029
kidsge6	.036005	.0434768	0.83	0.408	-.049208	.1212179
_cons	.2700768	.508593	0.53	0.595	-.7267473	1.266901
mills						
lambda	.0322619	.1336246	0.24	0.809	-.2296376	.2941613
rho	0.04861					
sigma	.66362875					

```
est clear
```

```
eststo OLS: reg lwage educ exper expersq
```

```
eststo Heckman: heckman lwage educ exper expersq, select(inlf=nwifeinc educ exper expersq age kidslt6 kidsge6)
```

```
esttab, mtitle
```

	(1) OLS	(2) Heckman
main		
educ	0.107*** (7.60)	0.109*** (7.03)
exper	0.0416** (3.15)	0.0439** (2.70)
expersq	-0.000811*	-0.000859

	(-2.06)	(-1.96)
_cons	-0.522** (-2.63)	-0.578 (-1.90)

inlf		
nwifeinc		-0.0120* (-2.48)
educ		0.131*** (5.18)
exper		0.123*** (6.59)
expersq		-0.00189** (-3.15)
age		-0.0529*** (-6.23)
kidslt6		-0.868*** (-7.33)
kidsge6		0.0360 (0.83)
_cons		0.270 (0.53)

mills		
lambda		0.0323 (0.24)

N	428	753

t statistics in parentheses		
* p<0.05, ** p<0.01, *** p<0.001		

There is no real evidence of sample selection bias in the wage offer equation. Our $\hat{\lambda}$ is not statistically significant and we fail to reject $H_0 : \rho = 0$. Also, we notice very little difference between our OLS and Heckman Method.

Interpretation: we will have a similar interpretation to our OLS. We have a log-linear model so our interpretation of the return would be e^β or $(e^\beta - 1) * 100$. Our $\hat{\lambda}$ would be a potential omitted variable, but our example shows that we

fail to reject the null hypothesis and selection bias is does not appear to be problematic.

Chapter 4

Subgroup Functions Part II

4.1 Computing values within subgroups: Running sums

As we mentioned earlier, when we use *egen sum* vs *gen sum*, we get different results. *Egen sum()* is similar to *egen total()*, but it can be confusing. When we use *gen* with *sum()*, we generate a **RUNNING** sum not a constant of total.

We can generate the tv running sum across all individuals over time

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"  
use tv1, clear  
generate tvrunsum = sum(tv)
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

We can generate the tv running sum within an individuals time period

```
bysort kidid (dt): generate bytvrunsum=sum(tv)
```

We can generate the total sum with an individuals time period

```
bysort kidid (dt): egen bytvsum=total(tv)
```

We can generate the total sum for all individuals over time

```
egen tvsum = total(tv)
```

We can also calculate a running average

```
bysort kidid (dt): generate bytvrnavg=sum(tv)/_n
```

We can compute the individual's average average

```
bysort kidid (dt): egen bytvavg = mean(tv)
list kidid tv tv* by*, sepby(kidid)
```

	kidid	tv	tv	tvrsum	tvsum	bytvr-m	bytvsum	bytvrun-g	bytvavg
1.	1	1	1	1	38	1	4	1	2
2.	1	3	3	4	38	4	4	2	2
3.	2	8	8	12	38	8	8	8	8
4.	3	2	2	14	38	2	14	2	3.5
5.	3	5	5	19	38	7	14	3.5	3.5
6.	3	1	1	20	38	8	14	2.6666667	3.5
7.	3	6	6	26	38	14	14	3.5	3.5
8.	4	7	7	33	38	7	12	7	4
9.	4	1	1	34	38	8	12	4	4
10.	4	4	4	38	38	12	12	4	4

4.2 Computing values within subgroups: More examples

There are other useful calculations we can do with subscripting/indexing. Some do overlap with egen, but it is helpful to know the differences.

Counting

Count the number of observations: this can be done with subscripting or egen depending upon what we want

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```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use tv1, clear

*Generate total observation count per individual missing or not missing:
bysort kidid (dt): generate idcount=_N

*Generate the total observation without missing
bysort kidid (dt): egen idcount_nomiss = count(tv)

*Generate a running count of an observation
bysort kidid (dt): gen idruncount = _n
list, sepby(kidid)
```

/Users/Sam/Desktop/Econ 645/Data/Mitchell

	kidid	dt	female	wt	tv	vac	idcount	idcount~s	idrunc~t
1.	1	07jan2002	1	53	1	1	2	2	1
2.	1	08jan2002	1	55	3	1	2	2	2
3.	2	16jan2002	1	58	8	1	1	1	1
4.	3	18jan2002	0	60	2	0	4	4	1
5.	3	19jan2002	0	63	5	1	4	4	2
6.	3	21jan2002	0	66	1	1	4	4	3
7.	3	22jan2002	0	64	6	0	4	4	4
8.	4	10jan2002	1	62	7	0	3	3	1
9.	4	11jan2002	1	58	1	0	3	3	2
10.	4	13jan2002	1	55	4	0	3	3	3

Generate Binaries

We can generate binary variables to find first and last observations or nth observation. This differs from id counts, we are generating binaries for when the qualifier is true.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use tv1, clear

*Find individuals with only one observation
bysort kidid (dt): generate singleob = (_N==1)
```

```

*Find the first observation of an individual
bysort kidid (dt): generate firstob = (_n==1)

*Find the last observation of an individual
bysort kidid (dt): generate lastob = (_n==_N)
list, sepby(kidid)

```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

	kidid	dt	female	wt	tv	vac	singleob	firstob	lastob
1.	1	07jan2002	1	53	1	1	0	1	0
2.	1	08jan2002	1	55	3	1	0	0	1
3.	2	16jan2002	1	58	8	1	1	1	1
4.	3	18jan2002	0	60	2	0	0	1	0
5.	3	19jan2002	0	63	5	1	0	0	0
6.	3	21jan2002	0	66	1	1	0	0	0
7.	3	22jan2002	0	64	6	0	0	0	1
8.	4	10jan2002	1	62	7	0	0	1	0
9.	4	11jan2002	1	58	1	0	0	0	0
10.	4	13jan2002	1	55	4	0	0	0	1

We can create binaries depending upon leads and lags too. Look for a change in vac

```

cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use tv1, clear
bysort kidid (dt): generate vacstart=(vac==1) & (vac[_n-1]==0)
bysort kidid (dt): generate vacend=(vac==1) & (vac[_n+1]==0)
list kidid dt vac*, sepby(kidid)

```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

	kidid	dt	vac	vacstart	vacend
1.	1	07jan2002	1	0	0
2.	1	08jan2002	1	0	0

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3.		2	16jan2002	1	0	0	

4.		3	18jan2002	0	0	0	
5.		3	19jan2002	1	1	0	
6.		3	21jan2002	1	0	1	
7.		3	22jan2002	0	0	0	

8.		4	10jan2002	0	0	0	
9.		4	11jan2002	0	0	0	
10.		4	13jan2002	0	0	0	

Fill in Missing

Another useful tool that we should use with caution is filling in missings. This should only really be applied when we have a constant variable that does not change over time.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use tv2, clear
sort kidid dt
list, sepby(kidid)
```

/Users/Sam/Desktop/Econ 645/Data/Mitchell

+-----+								
		kidid	dt	female	wt	tv	vac	
+-----+								
1.		1	07jan2002	1	53	1	1	
2.		1	08jan2002	1	55	3	1	
+-----+								
3.		2	16jan2002	1	58	8	1	
+-----+								
4.		3	18jan2002	0	60	2	0	
5.		3	19jan2002	0	.	.	.	
6.		3	21jan2002	0	66	.	1	
7.		3	22jan2002	0	64	6	0	
+-----+								
8.		4	10jan2002	1	62	7	0	
9.		4	11jan2002	1	58	.	.	
10.		4	13jan2002	1	.	4	0	
+-----+								

We can backfill the observation with the last nonmissing observation. First generate a copy of the variable with missing values.

```
generate tvimp1 = tv
bysort kidid (dt): replace tvimp1 = tv[_n-1] if missing(tv)
list kidid dt tv tvimp1, sepby(kidid)
```

(3 missing values generated)

(2 real changes made)

	kidid	dt	tv	tvimp1
1.	1	07jan2002	1	1
2.	1	08jan2002	3	3
3.	2	16jan2002	8	8
4.	3	18jan2002	2	2
5.	3	19jan2002	.	2
6.	3	21jan2002	.	.
7.	3	22jan2002	6	6
8.	4	10jan2002	7	7
9.	4	11jan2002	.	7
10.	4	13jan2002	4	4

Notice that we are still missing the 3rd observation for the 3rd kid. It cannot backfill the 3rd observation from the second observation, since the second observation is missing. There are a couple of strategies to use. We can generate a new variable like Mitchell.

```
generate tvimp2 = tvimp1
bysort kidid (dt): replace tvimp2 = tvimp2[_n-1] if missing(tvimp2)
list kidid tv tvimp*, sepby(kidid)
```

(1 missing value generated)

(1 real change made)

kidid	tv	tvimp1	tvimp2
1	1	1	1
1	3	3	3
2	8	8	8
3	2	2	2
3	.	2	2
3	.	.	.
3	6	6	6
4	7	7	7
4	.	7	7
4	4	4	4

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1.	1	1	1	1
2.	1	3	3	3
3.	2	8	8	8
4.	3	2	2	2
5.	3	.	2	2
6.	3	.	.	2
7.	3	6	6	6
8.	4	7	7	7
9.	4	.	7	7
10.	4	4	4	4

You can just replace tvimp1 twice instead of generating a new variable, but that is up to the user. You would use tv[_n-1] for the first replace and tvimp1[_n-1] for the second replace.

Interpolation

We may need to interpolate between 2 known values and assume a linear trend.

```
generate tvimp3=tv
*Interpolate for 1 missing value between two known values
bysort kidid (dt): replace tvimp3 = (tv[_n-1]+tv[_n+1])/2 if missing(tv)
list kidid tv tvimp3, sepby(kidid)
```

(3 missing values generated)

(1 real change made)

	kidid	tv	tvimp3
1.	1	1	1
2.	1	3	3
3.	2	8	8
4.	3	2	2
5.	3	.	.
6.	3	.	.

7.		3	6	6	

8.		4	7	7	
9.		4	.	5.5	
10.		4	4	4	

This is a bit of hard coding, but you can interpolate with more than 1 missing.

```
bysort kidid (dt): replace tvimp3 = ((tvimp3[4]-tvimp3[1])/3)+tvimp3[_n-1] if missing(
list kidid tv tvimp3, sepby(kidid)
```

(2 real changes made)

		kidid	tv	tvimp3	

1.		1	1	1	
2.		1	3	3	

3.		2	8	8	

4.		3	2	2	
5.		3	.	3.333333	
6.		3	.	4.666667	
7.		3	6	6	

8.		4	7	7	
9.		4	.	5.5	
10.		4	4	4	

Indicators

What is we want to find outliers in time-varying differences? We can generate indicators variables to find when a variable changes more than a set limit. For example we want to know if the tv viewing habits drop more than 2 hours

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use tv1, clear
bysort kidid (dt): generate tvchange = tv[_n]-tv[_n-1]
bysort kidid (dt): generate tvchangerate = ((tv[_n]-tv[_n-1])/tv[_n-1])
list, sepby(kidid)
```

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/Users/Sam/Desktop/Econ 645/Data/Mitchell

(4 missing values generated)

(4 missing values generated)

	kidid	dt	female	wt	tv	vac	tvchange	tvchange~e
1.	1	07jan2002	1	53	1	1	.	.
2.	1	08jan2002	1	55	3	1	2	2
3.	2	16jan2002	1	58	8	1	.	.
4.	3	18jan2002	0	60	2	0	.	.
5.	3	19jan2002	0	63	5	1	3	1.5
6.	3	21jan2002	0	66	1	1	-4	-.8
7.	3	22jan2002	0	64	6	0	5	5
8.	4	10jan2002	1	62	7	0	.	.
9.	4	11jan2002	1	58	1	0	-6	-.85714286
10.	4	13jan2002	1	55	4	0	3	3

Generate an indicator variable to see if tvchange is less than -2. This is not very helpful with small datasets, but it is important with larger datasets such as the CPS.

```
gen tvchangeid=(tvchange <= -2) if !missing(tvchange)
list, sepby(kidid)
```

(4 missing values generated)

	kidid	dt	female	wt	tv	vac	tvchange	tvchange~e	tvchan~d
1.	1	07jan2002	1	53	1	1	.	.	.
2.	1	08jan2002	1	55	3	1	2	2	0
3.	2	16jan2002	1	58	8	1	.	.	.
4.	3	18jan2002	0	60	2	0	.	.	.
5.	3	19jan2002	0	63	5	1	3	1.5	0
6.	3	21jan2002	0	66	1	1	-4	-.8	1

7.		3	22jan2002	0	64	6	0	5	5	0	

8.		4	10jan2002	1	62	7	0	.	.	.	
9.		4	11jan2002	1	58	1	0	-6	-.85714286	1	
10.		4	13jan2002	1	55	4	0	3	3	0	

4.3 Comparing the by, tsset, xtset commands

Another way to find differences within vectors, we can use the `tsset` or `xtset` command to establish the times series (`tsset`) or panel data (`xtset`). We can use our `bysort` with subscripting/indexing.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use tv1, clear
bysort kidid (dt): generate ltv = tv[_n-1]
list, sepby(kidid)
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

(4 missing values generated)

		kidid	dt	female	wt	tv	vac	ltv	

1.		1	07jan2002	1	53	1	1	.	
2.		1	08jan2002	1	55	3	1	1	

3.		2	16jan2002	1	58	8	1	.	

4.		3	18jan2002	0	60	2	0	.	
5.		3	19jan2002	0	63	5	1	2	
6.		3	21jan2002	0	66	1	1	5	
7.		3	22jan2002	0	64	6	0	1	

8.		4	10jan2002	1	62	7	0	.	
9.		4	11jan2002	1	58	1	0	7	
10.		4	13jan2002	1	55	4	0	1	

tsset

Instead we can establish a time series with `tsset`

We'll need to specify that our cross-sectional groups is kidid. We'll need to specify our date variable with dt. We'll need to use the option, daily, to specify that time period is daily as opposed to weeks, months, years. Or we can specify delta(1) for one day.

We can use the operator *L.var* to specify that we want to a lag of 1 day

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use tv1, clear
sort kidid dt
tsset kidid dt, daily delta(1)
generate lagtv = L.tv
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

```
panel variable: kidid (unbalanced)
time variable: dt, 07jan2002 to 22jan2002, but with gaps
delta: 1 day
```

```
(6 missing values generated)
```

We can use the operator *F.var* to specify that we want a lead of 1 day

```
generate leadtv = F.tv
list, sepby(kidid)
```

```
(6 missing values generated)
```

	kidid	dt	female	wt	tv	vac	lagtv	leadtv
1.	1	07jan2002	1	53	1	1	.	3
2.	1	08jan2002	1	55	3	1	1	.
3.	2	16jan2002	1	58	8	1	.	.
4.	3	18jan2002	0	60	2	0	.	5
5.	3	19jan2002	0	63	5	1	2	.
6.	3	21jan2002	0	66	1	1	.	6
7.	3	22jan2002	0	64	6	0	1	.
8.	4	10jan2002	1	62	7	0	.	1
9.	4	11jan2002	1	58	1	0	7	.
10.	4	13jan2002	1	55	4	0	.	.

xtset

We can establish a panel series with *xtset*

We'll need to specify that our cross-sectional group is kidid. We'll need to specify our time period is dt. We'll use a delta of 1 to specify that the difference is 1 day. Or, we can use *daily*, as well

Generate a lag with the l.var operator and generate a lead with the f.var

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use tv1, clear
sort kidid dt
xtset kidid dt, daily delta(1)
generate lagtv = l.tv
generate leadtv = f.tv
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

```
panel variable: kidid (unbalanced)
time variable: dt, 07jan2002 to 22jan2002, but with gaps
delta: 1 day
```

```
(6 missing values generated)
```

```
(6 missing values generated)
```

What do you notice? You can see that there are some leads and lags missing. Why? Because there is an unbalance panel and the daily differences cannot be computed if we are missing days. In this case, we can use the bysort with subscripting indexing

```
list, sepby(kidid)
bysort kidid (dt): gen bylagtv=tv[_n-1]
bysort kidid (dt): gen byleadtv=tv[_n+1]
list, sepby(kidid)
```

	kidid	dt	female	wt	tv	vac	lagtv	leadtv
1.	1	07jan2002	1	53	1	1	.	3
2.	1	08jan2002	1	55	3	1	1	.
3.	2	16jan2002	1	58	8	1	.	.
4.	3	18jan2002	0	60	2	0	.	5

5.		3	19jan2002	0	63	5	1	2	.	
6.		3	21jan2002	0	66	1	1	.	6	
7.		3	22jan2002	0	64	6	0	1	.	

8.		4	10jan2002	1	62	7	0	.	1	
9.		4	11jan2002	1	58	1	0	7	.	
10.		4	13jan2002	1	55	4	0	.	.	

(4 missing values generated)

(4 missing values generated)

		kidid	dt	female	wt	tv	vac	lagtv	leadtv	bylagtv	byleadtv	

1.		1	07jan2002	1	53	1	1	.	3	.	3	
2.		1	08jan2002	1	55	3	1	1	.	1	.	

3.		2	16jan2002	1	58	8	1	.	.	.	.	

4.		3	18jan2002	0	60	2	0	.	5	.	5	
5.		3	19jan2002	0	63	5	1	2	.	2	1	
6.		3	21jan2002	0	66	1	1	.	6	5	6	
7.		3	22jan2002	0	64	6	0	1	.	1	.	

8.		4	10jan2002	1	62	7	0	.	1	.	1	
9.		4	11jan2002	1	58	1	0	7	.	7	4	
10.		4	13jan2002	1	55	4	0	.	.	1	.	

What is the difference? xtset vs tsset

From Nick Cox: xtset allows a panel identifier only. tsset allows a time identifier only. Where they overlap is when two variables are supplied in which case the first is treated as a panel identifier and the second as a time variable.

4.4 Exercises

Let's grab the CPS and generate subgroup analysis. We will be using unweighted data for simplicity 1. What are the average wages by sex? 2. What are average wages by state 3. What are the median wages by sex 4. What are the median wages by state 5. What is the 75th percentile of wages by race? 6. What is the 25th percentile of wages by marital status?

```
use "/Users/Sam/Desktop/Econ 645/Data/CPS/jun23pub.dta",replace
```

Use bysort state: egen totalvar=total(var1)

Mitchell, N.M. (2020). *Data Management Using Stata: A Practical Handbook*. (2nd ed.). Stata Press.

UCLA Advanced Research Computing. (2025). *How can I access information stored after I run a command in Stata*. Retrieved from <https://stats.oarc.ucla.edu/stata/faq/how-can-i-access-information-stored-after-i-run-a-command-in-stata-returned-results/>

Wooldridge (2009). *Introductory Econometrics: A Modern Approach*. (4th ed.). South-Western Cengage Learning.